Task 1 — Open the router login page

Procedure: I first connected my laptop/mobile to my home Wi-Fi. On Windows, I would open Command Prompt and type `ipconfig`. I would look for the **Default Gateway** value, then type that address in the browser address bar. Common examples are 192.168.0.1 and 192.168.1.1, but the address depends on the router.

Browser address	http://192.168.1.1
Router Admin	Sign in to manage your Wi-Fi network
Username	Enter router username
Password	Enter router password
	[Log In]

Illustration only — this is a sample login-page layout, not a screenshot of your actual router. Use your own router's page for the real screenshot in your practical.

Observation: The login page asks for administrator credentials. These are different from the Wi-Fi password on some routers. If the login details are unknown, check the router label or the ISP/router documentation rather than guessing repeatedly.

Task 2 — Note the device's IP details

I would check the values while connected to my Wi-Fi. On Windows, I can run `ipconfig` in Command Prompt. On a phone, I can open the connected Wi-Fi network's details. The table below shows an example format; it is not a claim about my actual network.

Setting	Example value	Meaning
IPv4 address	192.168.1.25	Address assigned to this device
Subnet mask	255.255.255.0	Shows which addresses are on the local network
Default gateway	192.168.1.1	Usually the router address used to reach other networks

Windows command:

```
ipconfig
```

I would copy the IPv4 Address, Subnet Mask and Default Gateway values from the Wi-Fi adapter section. If the device is connected to mobile data instead of Wi-Fi, these values may not represent the home router network.

Task 3 — Change DHCP to a static IP and test

A static IP is an address entered manually instead of being assigned automatically by DHCP. For a safe classroom test, I would first record the original settings and use an unused address in the same subnet. The example below assumes the router is 192.168.1.1 and the subnet mask is 255.255.255.0.

Setting	Example for test only
IP address	192.168.1.50
Subnet mask	255.255.255.0
Default gateway	192.168.1.1
DNS server	Use the router address or a DNS server approved for the network

Steps:

• Open the network adapter settings on the device and choose IPv4 properties.

• Change from 'Obtain an IP address automatically' to manual entry, then enter values appropriate to the real network.

• Reconnect or test access to the router and internet. If internet access stops, check the values and restore automatic/DHCP settings.

• After the test, select 'Obtain an IP address automatically' and automatic DNS again, unless the network administrator has instructed otherwise.

Why must I revert? A manually selected address can conflict with another device or be outside the router's intended settings. Returning to DHCP lets the router assign an address automatically and reduces the chance of a network problem.

Important: Do not use 192.168.1.50 blindly on a real network. Confirm the router's subnet and make sure the address is not already in use. On managed/college networks, ask the administrator before changing IP settings.

Task 4 — Find the DHCP range and explain duplicate IPs

In the router admin page, I would look under a menu such as *Network / LAN / DHCP Server*. The exact menu name changes by router brand. The DHCP range is the group of addresses the router can automatically give to connected devices.

Item	Example value
DHCP start address	192.168.1.100
DHCP end address	192.168.1.199
Address pool	192.168.1.100–192.168.1.199

What happens if two devices have the same IP address?

If two devices use the same IP address on one network, an IP conflict can occur. One or both devices may lose reliable network access, and connections can become intermittent because the network cannot consistently identify the correct device. A router's DHCP service normally avoids duplicate assignments

within its managed pool, but manually configured addresses can still conflict if they overlap with addresses already in use.